

KYOWOON LEE

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RESEARCH INTERESTS

My long-term research vision centers on developing trustworthy and scalable decision-making systems. To this end, I aim to: (1) develop data-efficient reinforcement learning algorithms that can learn effectively even in data-scarce or costly-to-simulate environments, addressing the fundamental challenge of limited data availability in real-world applications; and (2) ensure that autonomous agents make reliable, safe, and human-aligned decisions that can be trusted and seamlessly integrated into complex real-world systems such as robotics and interactive environments.

PROFESSIONAL EXPERIENCE

Ajou University

Assistant Professor, Department of Mobility Engineering

Sep 2026 - present

Korea Advanced Institute of Science and Technology (KAIST)

Postdoctoral Researcher

Oct 2024 - Aug 2026

EDUCATION

Ulsan National Institute of Science and Technology (UNIST)

Combined M.S. and Ph.D. Program in Computer Science and Engineering

Sep 2016 - Aug 2024

Ulsan National Institute of Science and Technology (UNIST)

B.S. in Computer Science and Engineering, *summa cum laude*

Mar 2012 - Aug 2016

GPA (Overall): 4.01/4.3

PUBLICATIONS

International Conferences (*: equal contribution, †: equal advising)

1. Soyeon Kim, **Kywoon Lee**[†] and Jaesik Choi[†], *Diffusion Integrated Gradients: Controllable Path Generation for Flexible Feature Attribution*, European Conference on Computer Vision (**ECCV**), 2026.
2. Soyeon Kim, Seongwoo Lim, **Kywoon Lee**[†] and Jaesik Choi[†], *Spectral Integrated Gradients for Coarse-to-Fine Feature Attribution*, ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2026.
3. Soyeon Kim, Seongwoo Lim, **Kywoon Lee**[†] and Jaesik Choi[†], *Manifold-Aligned Guided Integrated Gradients for Reliable Feature Attribution*, International Conference on Machine Learning (**ICML**), 2026.
4. **Kywoon Lee** and Jaesik Choi, *State-Covering Trajectory Stitching for Diffusion Planning*, Conference on Neural Information Processing Systems (**NeurIPS**), 2025.
5. **Kywoon Lee** and Jaesik Choi, *Local Manifold Approximation and Projection for Manifold-Aware Diffusion Planning*, International Conference on Machine Learning (**ICML**), 2025.
6. **Kywoon Lee**, Artyom Stitsyuk, Gunu Jho, Inchul Hwang and Jaesik Choi, *Counterfactual Activation Editing for Post-hoc Prosody and Mispronunciation Correction in TTS Models*, **Inter-speech**, 2025.
7. **Kywoon Lee**^{*}, Seongun Kim^{*} and Jaesik Choi, *Refining Diffusion Planner for Reliable Behavior Synthesis by Automatic Detection of Infeasible Plans*, Conference on Neural Information Processing Systems (**NeurIPS**), 2023.

8. Seongun Kim*, **Kywoon Lee*** and Jaesik Choi, *Variational Curriculum Reinforcement Learning for Unsupervised Discovery of Skills*, International Conference on Machine Learning (ICML), 2023.
9. **Kywoon Lee***, Seongun Kim* and Jaesik Choi, *Adaptive and Explainable Deployment of Navigation Skills via Hierarchical Deep Reinforcement Learning*, International Conference on Robotics and Automation (ICRA), 2023.
10. Jiyeon Han*, **Kywoon Lee***, Anh Tong and Jaesik Choi, *Confirmatory Bayesian Online Change Point Detection in the Covariance Structure of Gaussian Processes*, International Joint Conference on Artificial Intelligence (IJCAI), 2019.
11. **Kywoon Lee***, Sol-A Kim*, Jaesik Choi and Seong-Hwan Lee, *Deep Reinforcement Learning in Continuous Action Spaces: a Case Study in the Game of Simulated Curling*, International Conference on Machine Learning (ICML), 2018.

WORKING PAPERS

Under Review (*: equal contribution, †: equal advising)

1. **Kywoon Lee**, Yunhao Luo, Anh Tong and Jaesik Choi, *Refining Compositional Diffusion for Reliable Long-Horizon Planning*.
2. Soyeon Kim*, Seongwoo Lim*, **Kywoon Lee** and Jaesik Choi, *Robust Path Attribution in Piece-wise Linear Neural Networks via Dummy-Constrained Optimization*.

PATENTS

1. Jaesik Choi, Jiyeon Han and **Kywoon Lee**, *Methods and Apparatus for Predicting Data*, Conference on Neural Information Processing Systems, KR Patent 10-2020-0110650, 2020.
2. Jaesik Choi, Sol-A Kim and **Kywoon Lee**, *Apparatus for Presenting Pitching Strategy of Curling Game Stone and Operation Method of the Apparatus*, Conference on Neural Information Processing Systems, KR Patent 10-2017-0151643, 2017.

HONORS AND AWARDS

- Compute Transparency Champion, CVPR 2026.
- Silver Reviewer, ICML 2026.
- InnoCORE Postdoctoral Fellowship, Ministry of Science and ICT (MSIT), 2025–2026.
- Top Reviewer, NeurIPS 2025.
- Naver Ph.D. Fellowship Award, Naver, 2018.
- National Science and Technology Scholarship, Korean Student Aid Foundation, 2012–2016.
- SAIL Research Excellence Award, Statistical Artificial Intelligence Lab, UNIST, 2018.
- Summa Cum Laude, UNIST, 2016.
- **Winner (the 1st place)**, Breast Cancer Classification on Frozen Pathology, HeLP Challenge at Asan Medical Center, 2019.
- **Winner (the 1st place)**, UEC-cup Digital Curling Competition, Game AI Tournament, 2018.
- **Winner (the 1st place)**, Digital Curling Competition, Game Playing Workshop, 2017.

ACADEMIC SERVICES

Reviewer: NeurIPS, ICML, ICLR, RSS, IROS, CVPR, ECCV, KDD.

REFERENCES

Prof. Jaesik Choi: Professor in the Graduate School of AI, KAIST